

ANDREW ZENG

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EDUCATION

Northeastern University

Boston, MA

B.S. in Electrical Engineering, Minor in Computer Science

May 2028

GPA: 3.29 | **Honors:** NU College of Engineering Merit Scholarship, Global Scholars Program

Activities: NU Lunabotics, NU SEDS, Google Developer Group, Club Golf, NU Downhillers, Club Spikeball

Relevant Coursework: Embedded Design, Discrete Structures, Fundamentals of CS II, Computing for Engineers

SKILLS

Technical & Embedded: Python, MATLAB, C++, JavaScript, Arduino, PCB Design (KiCad), RF Signal Processing

Design, Fabrication & Tools: SolidWorks, AutoCAD, Circuit Design, Soldering, 3D Printing, VS Code, Jupyter

EXPERIENCE

Westinghouse Electric Corporation

Pittsburgh, PA

Incoming Electrical Engineering Intern (OEM Parts Electrical Division)

Apr 2026

- Provide electrical system design & wiring, instrumentation and controls, and technical analysis for nuclear projects

University of Michigan

Ann Arbor, MI

Research Intern

May 2025 – Present

- Interpret sensor data with Python machine learning systems on custom PCBs for autonomous driving vision
- Develop Python-Manim visualizations for university lectures on core automotive electrical engineering topics
- Integrate advanced perception and control systems in RC car platforms illustrating autonomous driving concepts

Google Developer Group

Oakland, CA

Lead Partnerships Coordinator

Aug 2024 – May 2025

- Initiated and maintained relationships with 50+ tech companies and partners to foster collaboration and growth
- Curated and executed meetings, development sessions, and networking events for over 500 freshmen on campus
- Led a team of partnership coordinators to maintain partner engagement, and streamline collaboration

PROJECTS AND PUBLICATIONS

FMCW Radar

Feb 2026 - Present

- Design and implement FMCW mmWave radar system, capturing raw IQ data and processing through DSP pipeline
- Develop and validate signal processing algorithms (FFT-based, Doppler, detection, target tracking) using Python
- Integrate end-to-end hardware and software producing demos, technical documentation, and radar visualizations

Custom Bus PCB (Lunabotics)

Jan 2026 - Present

- Fabricate custom bus PCB using I²C communication system to coordinate sensor boards in lunar rover system
- Create schematics in KiCAD, including signal routing, connectors, and interface logic for board communication
- Implement power management architecture, distributing power to sensor components ensuring correct output

Self-Driving RC Car

May 2024 - Aug 2024

- Designed in AutoCAD and fabricated 3D-printed and laser-cut car utilizing Arduino Nano hardware and sensors
- Achieved autonomous navigation through Python-trained ML models with live ultrasonic and IR sensor feedback
- Developed vehicle models using Python, integrating Arduino-based feedback and AutoCAD chassis components

“DEMONSTRATION OF THE HYDROGEN CYCLE USING A PEM FUEL CELL”

by Andrew Zeng, Devon Renock- PhD

May 2023 - Aug 2023

- Researched Fuel Cells and increased efficiency by 15% using membranes containing higher proton conductivity
- Published findings as the only high school presenter at the 2023 American Chemical Society Fall Conference
- Granted sponsor award from USAF and Point Park University Scholarship worth ~\$30,000 yearly at PRSEF 2023